

Data Privacy: FinalSolutions

1. Nonuniform Time Scale Diffusion (30 points)

For a data variable X , consider a diffusion model with latent variables Z_t , where t runs from $t = 0$ (least noisy) to $t = 1$ (most noisy). The distribution of latent variable Z_t conditioned on X for $t \in [0, 1]$ is given by

$$q(Z_t|X) \sim \mathcal{N}(\alpha_t X, \sigma_t^2).$$

Furthermore, for $t > s$, the conditional distribution Z_t given Z_s is given by

$$q(Z_t|Z_s) \sim \mathcal{N}(\alpha_{t|s} Z_s, \sigma_{t|s}^2)$$

where $\alpha_{t|s} = \alpha_t / \alpha_s$ and $\sigma_{t|s}^2 = \sigma_t^2 - \alpha_{t|s}^2 \sigma_s^2$.

(a) Let $q(Z_s|Z_t, X) \sim \mathcal{N}(\mu_Q(Z_t, X; s, t), \sigma_Q^2(s, t))$. Find $\mu_Q(Z_t, X; s, t)$ and $\sigma_Q^2(s, t)$. (10 points)

(b) Let $t_i = i/T$ for $0 \leq i \leq T$. Consider a following reverse distribution p :

$$p(X) = \int_{\mathbf{Z}} p(Z_1) p(X|Z_0) \prod_{i=1}^T p(Z_{t_{i-1}}|Z_{t_i}) d\mathbf{Z}$$

where Z represents the whole $(Z_{t_0}, \dots, Z_{t_T})$.

Show the following variational bound

$$-\log p(X) \leq D(q(Z_1|X) \| p(Z_1)) + \mathbb{E}_{q(Z_0|X)}[-\log p(X|Z_0)] + \mathcal{L}(X)$$

where

$$\mathcal{L}(X) = \sum_{i=1}^T \mathbb{E}_{q(Z_{t_i}|X)} [D(q(Z_{t_{i-1}}|Z_{t_i}, X) \| p(Z_{t_{i-1}}|Z_{t_i}))].$$

(10 points)

(c) Let $p(Z_{t_{i-1}}|Z_{t_i}) = q(Z_{t_{i-1}}|Z_{t_i}, X = \hat{x}_\theta(Z_{t_i}; t_i))$. Show that

$$\mathcal{L}(X) = \frac{1}{2} \sum_{i=1}^T \mathbb{E}_{q(Z_{t_i}|X)} [(\text{SNR}(t_{i-1}) - \text{SNR}(t_i)) \|X - \hat{x}_\theta(Z_{t_i}; t_i)\|^2]$$

where $\text{SNR}(t) = \alpha_t^2 / \sigma_t^2$. (10 points)

Solution: Nonuniform Time Scale Diffusion.

(a) Since

$$\begin{bmatrix} Z_s \\ Z_t \end{bmatrix} \Big| X \sim \mathcal{N} \left(\begin{bmatrix} \alpha_s X \\ \alpha_t X \end{bmatrix}, \begin{bmatrix} \sigma_s^2 & \alpha_{t|s} \sigma_s^2 \\ \alpha_{t|s} \sigma_s^2 & \sigma_t^2 \end{bmatrix} \right),$$

the conditional distribution is given by

$$Z_s | Z_t, X \sim \mathcal{N} \left(\alpha_s X + \frac{\alpha_{t|s} \sigma_s^2}{\sigma_t^2} (Z_t - \alpha_t X), \sigma_s^2 - \frac{(\alpha_{t|s} \sigma_s^2)^2}{\sigma_t^2} \right).$$

Thus,

$$\begin{aligned}
\mu_Q(Z_t, X; s, t) &= \alpha_s X + \frac{\alpha_{t|s} \sigma_s^2}{\sigma_t^2} (Z_t - \alpha_t X) \\
&= \frac{\alpha_{t|s} \sigma_s^2}{\sigma_t^2} Z_t + \alpha_s X - \frac{\alpha_{t|s} \sigma_s^2}{\sigma_t^2} (\alpha_{t|s} \alpha_s) X \\
&= \frac{\alpha_{t|s} \sigma_s^2}{\sigma_t^2} Z_t + \alpha_s \left(1 - \frac{\alpha_{t|s}^2 \sigma_s^2}{\sigma_t^2} \right) X \\
&= \frac{\alpha_{t|s} \sigma_s^2}{\sigma_t^2} Z_t + \frac{\alpha_s \sigma_{t|s}^2}{\sigma_t^2} X,
\end{aligned}$$

and

$$\sigma_Q^2(s, t) = \sigma_s^2 - \frac{(\alpha_{t|s} \sigma_s^2)^2}{\sigma_t^2} = \frac{\sigma_s^2 \sigma_{t|s}^2}{\sigma_t^2}.$$

(b) First, we have

$$\begin{aligned}
-\log p(X) &= -\log \int p(X, \mathbf{Z}) d\mathbf{Z} \\
&= -\log \int q(\mathbf{Z}|X) \frac{p(X, \mathbf{Z})}{q(\mathbf{Z}|X)} d\mathbf{Z} \\
&= -\log \mathbb{E}_{q(\mathbf{Z}|X)} \left[\frac{p(X, \mathbf{Z})}{q(\mathbf{Z}|X)} \right] \\
&\leq \mathbb{E}_{q(\mathbf{Z}|X)} \left[\log \frac{q(\mathbf{Z}|X)}{p(X, \mathbf{Z})} \right]
\end{aligned}$$

by Jensen's inequality. Now we have factorizations

$$p(X, \mathbf{Z}) = p(X, Z_{t_0}, \dots, Z_{t_T}) = p(Z_1) p(X|Z_0) \prod_{i=1}^T p(Z_{t_{i-1}}|Z_{t_i})$$

and

$$q(\mathbf{Z}|X) = q(Z_{t_0}, \dots, Z_{t_T}|X) = q(Z_0|X) \prod_{i=1}^T q(Z_{t_i}|Z_{t_{i-1}}, X).$$

By Bayes' theorem,

$$q(Z_{t_i}|Z_{t_{i-1}}, X) = q(Z_{t_i}|Z_{t_{i-1}}) = \frac{q(Z_{t_{i-1}}|Z_{t_i}, X) q(Z_{t_i}|X)}{q(Z_{t_{i-1}}|X)}.$$

Then,

$$\begin{aligned}
& \mathbb{E}_{q(\mathbf{Z}|X)} \left[\log \frac{q(\mathbf{Z}|X)}{p(X, \mathbf{Z})} \right] \\
&= \mathbb{E}_{q(\mathbf{Z}|X)} \left[\log \frac{q(Z_0|X) \prod_{i=1}^T q(Z_{t_i}|Z_{t_{i-1}})}{p(Z_1)p(X|Z_0) \prod_{i=1}^T p(Z_{t_{i-1}}|Z_{t_i})} \right] \\
&= \mathbb{E}_{q(\mathbf{Z}|X)} \left[\log q(Z_0|X) - \log p(X|Z_0) - \log p(Z_1) \right. \\
&\quad \left. + \sum_{i=1}^T (\log q(Z_{t_i}|Z_{t_{i-1}}) - \log p(Z_{t_{i-1}}|Z_{t_i})) \right] \\
&= \mathbb{E}_{q(\mathbf{Z}|X)} \left[\log q(Z_0|X) - \log p(X|Z_0) - \log p(Z_1) \right. \\
&\quad \left. + \sum_{i=1}^T \left(\log \frac{q(Z_{t_{i-1}}|Z_{t_i}, X)q(Z_{t_i}|X)}{q(Z_{t_{i-1}}|X)} - \log p(Z_{t_{i-1}}|Z_{t_i}) \right) \right] \\
&= \mathbb{E}_{q(\mathbf{Z}|X)} \left[\log q(Z_0|X) - \log p(X|Z_0) - \log p(Z_1) \right. \\
&\quad \left. + \sum_{i=1}^T (\log q(Z_{t_i}|X) - \log q(Z_{t_{i-1}}|X)) + \sum_{i=1}^T \log \frac{q(Z_{t_{i-1}}|Z_{t_i}, X)}{p(Z_{t_{i-1}}|Z_{t_i})} \right] \\
&= \mathbb{E}_{q(\mathbf{Z}|X)} \left[\log q(Z_0|X) - \log p(X|Z_0) - \log p(Z_1) \right. \\
&\quad \left. + (\log q(Z_1|X) - \log q(Z_0|X)) + \sum_{i=1}^T \log \frac{q(Z_{t_{i-1}}|Z_{t_i}, X)}{p(Z_{t_{i-1}}|Z_{t_i})} \right] \\
&= \mathbb{E}_{q(Z_0|X)} [-\log p(X|Z_0)] + D(q(Z_1|X) \| p(Z_1)) + \mathcal{L}(X).
\end{aligned}$$

(c) We have

$$p(Z_{t_{i-1}}|Z_{t_i}) \sim \mathcal{N}(\mu_\theta(Z_{t_i}; t_{i-1}, t_i), \sigma_Q^2(t_{i-1}, t_i)),$$

where $\mu_\theta(Z_{t_i}; t_{i-1}, t_i) = \frac{\alpha_{t_i|t_{i-1}} \sigma_{t_{i-1}}^2}{\sigma_{t_i}^2} Z_t + \frac{\alpha_{t_{i-1}} \sigma_{t_i|t_{i-1}}^2}{\sigma_{t_i}^2} \hat{x}_\theta(Z_{t_i}; t_i)$. The KL divergence between two Gaussian distributions with common variance is given by:

$$\begin{aligned}
D(q(Z_{t_{i-1}}|Z_{t_i}, X) \| p(Z_{t_{i-1}}|Z_{t_i})) &= \frac{1}{2\sigma_Q^2(t_{i-1}, t_i)} \|\mu_Q - \mu_\theta\|^2 \\
&= \frac{\sigma_{t_i}^2}{2\sigma_{t_i|t_{i-1}}^2 \sigma_{t_{i-1}}^2} \frac{\alpha_{t_{i-1}}^2 \sigma_{t_i|t_{i-1}}^4}{\sigma_{t_i}^4} \|X - \hat{x}_\theta(Z_{t_i}; t_i)\|^2 \\
&= \frac{1}{2\sigma_{t_{i-1}}^2} \frac{\alpha_{t_{i-1}}^2 \sigma_{t_i|t_{i-1}}^2}{\sigma_{t_i}^2} \|X - \hat{x}_\theta(Z_{t_i}; t_i)\|^2 \\
&= \frac{1}{2\sigma_{t_{i-1}}^2} \frac{\alpha_{t_{i-1}}^2 (\sigma_{t_i}^2 - \alpha_{t_i|t_{i-1}}^2 \sigma_{t_{i-1}}^2)}{\sigma_{t_i}^2} \|X - \hat{x}_\theta(Z_{t_i}; t_i)\|^2 \\
&= \frac{1}{2} \frac{\alpha_{t_{i-1}}^2 \sigma_{t_i}^2 / \sigma_{t_{i-1}}^2 - \alpha_{t_i}^2}{\sigma_{t_i}^2} \|X - \hat{x}_\theta(Z_{t_i}; t_i)\|^2 \\
&= \frac{1}{2} (\text{SNR}(t_{i-1}) - \text{SNR}(t_i)) \|X - \hat{x}_\theta(Z_{t_i}; t_i)\|^2.
\end{aligned}$$

Thus,

$$\begin{aligned}\mathcal{L}(X) &= \sum_{i=1}^T \mathbb{E}_{q(Z_{t_i}|X)} [D(q(Z_{t_{i-1}}|Z_{t_i}, X) \| p(Z_{t_{i-1}}|Z_{t_i}))] \\ &= \frac{1}{2} \sum_{i=1}^T \mathbb{E}_{q(Z_{t_i}|X)} [(\text{SNR}(t_{i-1}) - \text{SNR}(t_i)) \|X - \hat{x}_\theta(Z_{t_i}; t_i)\|^2].\end{aligned}$$

2. Masked Diffusion (30 points)

Consider a masked discrete diffusion model with absorbing matrix. More precisely, we have $X_0 \sim p_{\text{data}}$ and the forward process $X_1, X_2, \dots, X_n$ where $X_i \sim p_i$ with $p_i = Q_{\text{absorb}} p_{i-1}$ for all i . Note that

$$Q_{\text{absorb}} = \begin{bmatrix} 1 - \epsilon & 0 & \dots & 0 & 0 \\ 0 & 1 - \epsilon & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 1 - \epsilon & 0 \\ \epsilon & \epsilon & \dots & \epsilon & 1 \end{bmatrix}$$

- (a) Find $p(X_i = x | X_0 = x)$ as a function of i and ϵ . (10 points)
- (b) Let $Q^{(i)}$ be a transition matrix that satisfies $p_i = Q^{(i)} p_0$. Find $Q^{(i)}$. (10 points)
- (c) We re-parametrize the forward process $\{Y_t\}_{t=0}^T$ so that the Y_t is now has continuous time scale. More precisely, we have $Y_{i\epsilon} = X_i$ where $T = n\epsilon$. Let q_t be the distribution of Y_t (i.e., $q_{i\epsilon} = p_i$). We still have $q_{t+\epsilon} = Q_{\text{absorb}} q_t$. Find Q_t that satisfies

$$\frac{1}{\epsilon}(q_{t+\epsilon} - q_t) = Q_t q_t.$$

(10 points)

Solution: Masked Diffusion.

- (a) Let $\mathcal{X} = \{1, 2, \dots, K\}$ be the state space and let $\overline{\mathcal{X}} = \mathcal{X} \cup \{[\mathbf{M}]\}$, so that $Q_{\text{absorb}} \in \mathbb{R}^{(K+1) \times (K+1)}$. Then, the transition probability at each step is given by:

$$p_{i|i-1}(y|x) = \begin{cases} 1 - \epsilon & \text{if } y = x \neq [\mathbf{M}] \\ \epsilon & \text{if } x \neq y = [\mathbf{M}] \\ 0 & \text{otherwise} \end{cases}$$

Notably, transitions between two distinct unmasked tokens and from the masked token to unmasked tokens are not allowed.

By the Markov property of the diffusion forward process, we have:

$$p_{i|0}(x|x) = \sum_{x_{i-1} \in \overline{\mathcal{X}}} \sum_{x_{i-2} \in \overline{\mathcal{X}}} \dots \sum_{x_1 \in \overline{\mathcal{X}}} p_{i|i-1}(x|x_{i-1}) p_{i-1|i-2}(x_{i-1}|x_{i-2}) \dots p_{1|0}(x_1|x)$$

Taking into account the disallowed transitions, we obtain:

$$\begin{aligned} p_{i|0}(x|x) &= p_{i|i-1}(x|x)p_{i-1|i-2}(x|x) \cdots p_{1|0}(x|x) \\ &= (1 - \epsilon)(1 - \epsilon) \cdots (1 - \epsilon) \\ &= (1 - \epsilon)^i. \end{aligned}$$

- (b) Observe that $Q^{(i)}(y, x) = p_{i|0}(y|x)$. Again, considering the disallowed transitions, for $y \neq x$ (both unmasked), we have $p_{i|0}(y|x) = 0$. From the previous result, $p_{i|0}(x|x) = (1 - \epsilon)^i$. Hence,

$$\begin{aligned} p_{i|0}([M]|x) &= 1 - \sum_{y=1}^K p_{i|0}(y|x) \\ &= 1 - p_{i|0}(x|x) \\ &= 1 - (1 - \epsilon)^i. \end{aligned}$$

Therefore, the matrix $Q^{(i)}$ is:

$$Q^{(i)} = \begin{bmatrix} (1 - \epsilon)^i & 0 & \cdots & 0 & 0 \\ 0 & (1 - \epsilon)^i & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & (1 - \epsilon)^i & 0 \\ 1 - (1 - \epsilon)^i & 1 - (1 - \epsilon)^i & \cdots & 1 - (1 - \epsilon)^i & 1 \end{bmatrix}.$$

Alternative approach. Alternatively, $Q^{(i)}$ can be derived by computing the i -th power of the matrix Q_{absorb} . This power can be computed either by induction or by using the decomposition $Q_{\text{absorb}} = (1 - \epsilon)I + \epsilon A$ together with the binomial expansion.

(c)

$$\begin{aligned} q_{t+\epsilon} - q_t &= Q_{\text{absorb}} q_t - q_t \\ &= (Q_{\text{absorb}} - I) q_t, \end{aligned}$$

where I is the identity matrix. Thus,

$$\begin{aligned} Q_t &= \frac{1}{\epsilon} (Q_{\text{absorb}} - I) \\ &= \frac{1}{\epsilon} \begin{bmatrix} -\epsilon & 0 & \cdots & 0 & 0 \\ 0 & -\epsilon & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & -\epsilon & 0 \\ \epsilon & \epsilon & \cdots & \epsilon & 0 \end{bmatrix} = \begin{bmatrix} -1 & 0 & \cdots & 0 & 0 \\ 0 & -1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & -1 & 0 \\ 1 & 1 & \cdots & 1 & 0 \end{bmatrix}. \end{aligned}$$

3. Discrete Langevin (40 points)

In the homework problem, we showed that the Langevin SDE

$$dX_t = \frac{1}{2} \nabla_{X_t} \log p(X_t) dt + dW_t.$$

preserves the distribution p for all t , if the initial distribution satisfies $p_0 = p$. However, this result can be significantly strengthened. Specifically, even when the initial distribution p_0 differs from p , the distribution of $X_t \sim p_t$ will converge to the target distribution p as $t \rightarrow \infty$.

In this problem, we examine this convergence behavior in a simple setting where the target distribution p is Gaussian. Consider the probability density function of the Gaussian distribution $\mathcal{N}(\mu, \sigma^2)$:

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

Let this Gaussian $p(x)$ be our target distribution.

- (a) Compute the score function $\frac{\partial}{\partial x} \log p(x)$. (10 points)
- (b) Consider the discretized Langevin SDE with a small ϵ :

$$X_{i+1} = X_i + \epsilon \cdot \frac{\partial}{\partial x} \log p(X_i) + \sqrt{2\epsilon} Z_i$$

where $Z_i \sim \mathcal{N}(0, 1)$ is an independent Gaussian noise. Suppose the initial distribution is also standard Gaussian $X_0 \sim \mathcal{N}(0, 1)$, which differs from the target distribution p . Show that X_i is Gaussian for all i . (10 points)

- (c) Our goal is to show that p_{X_n} converges to p as n grows. Since X_i is Gaussian, we can write $X_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$. Derive the recursive relations for μ_{i+1} in terms of μ_i , and σ_{i+1}^2 in terms of σ_i^2 . Show that $\mu_n \rightarrow \mu$ as $n \rightarrow \infty$. (10 points)
- (d) Using a suitable approximation, show that $\sigma_n^2 \rightarrow \sigma^2$ as $n \rightarrow \infty$, assuming ϵ is small. (10 points)

Solution: Discrete Langevin.

- (a) The score function is simply

$$\frac{\partial}{\partial x} \log p(x) = -\frac{x - \mu}{\sigma^2}.$$

- (b) Using the above form of the score function, the discretized Langevin SDE is

$$\begin{aligned} X_{i+1} &= X_i + \epsilon \cdot \frac{X_i - \mu}{\sigma^2} + \sqrt{2\epsilon} Z_i \\ &= \left(1 - \frac{\epsilon}{\sigma^2}\right) X_i + \frac{\epsilon}{\sigma^2} \mu + \sqrt{2\epsilon} Z_i \end{aligned}$$

Since $\epsilon\mu/\sigma^2$ is a constant, X_{i+1} is a linear combination of X_i and Z_i . Thus, if X_i is Gaussian, then X_{i+1} is also Gaussian.

- (c) Using the above recursive relation, we have

$$\mu_{i+1} = \left(1 - \frac{\epsilon}{\sigma^2}\right) \mu_i + \frac{\epsilon}{\sigma^2} \mu$$

which is a convex combination of μ_i and μ . Thus, if we iteratively apply this relation, we have $\mu_i \rightarrow \mu$. Or simply, we can take the limit to both sides, we have

$$\lim \mu_n = \left(1 - \frac{\epsilon}{\sigma^2}\right) \lim \mu_n + \frac{\epsilon}{\sigma^2} \mu$$

which implies $\lim \mu_n = \mu$.

Similarly, we have

$$\sigma_{i+1}^2 = \left(1 - \frac{\epsilon}{\sigma^2}\right)^2 \sigma_i^2 + 2\epsilon.$$

(d) If we take the limit to both sides, we have

$$\lim \sigma_n^2 = \left(1 - \frac{\epsilon}{\sigma^2}\right)^2 \lim \sigma_n^2 + 2\epsilon.$$

which implies

$$\begin{aligned} \lim \sigma_n^2 &= \frac{2\epsilon}{1 - (1 - \epsilon/\sigma^2)^2} \\ &\approx \frac{2\epsilon}{1 - (1 - 2\epsilon/\sigma^2)} \\ &= \sigma^2. \end{aligned}$$

Note that the approximation is valid when ϵ is small.